

Practical Work Plan

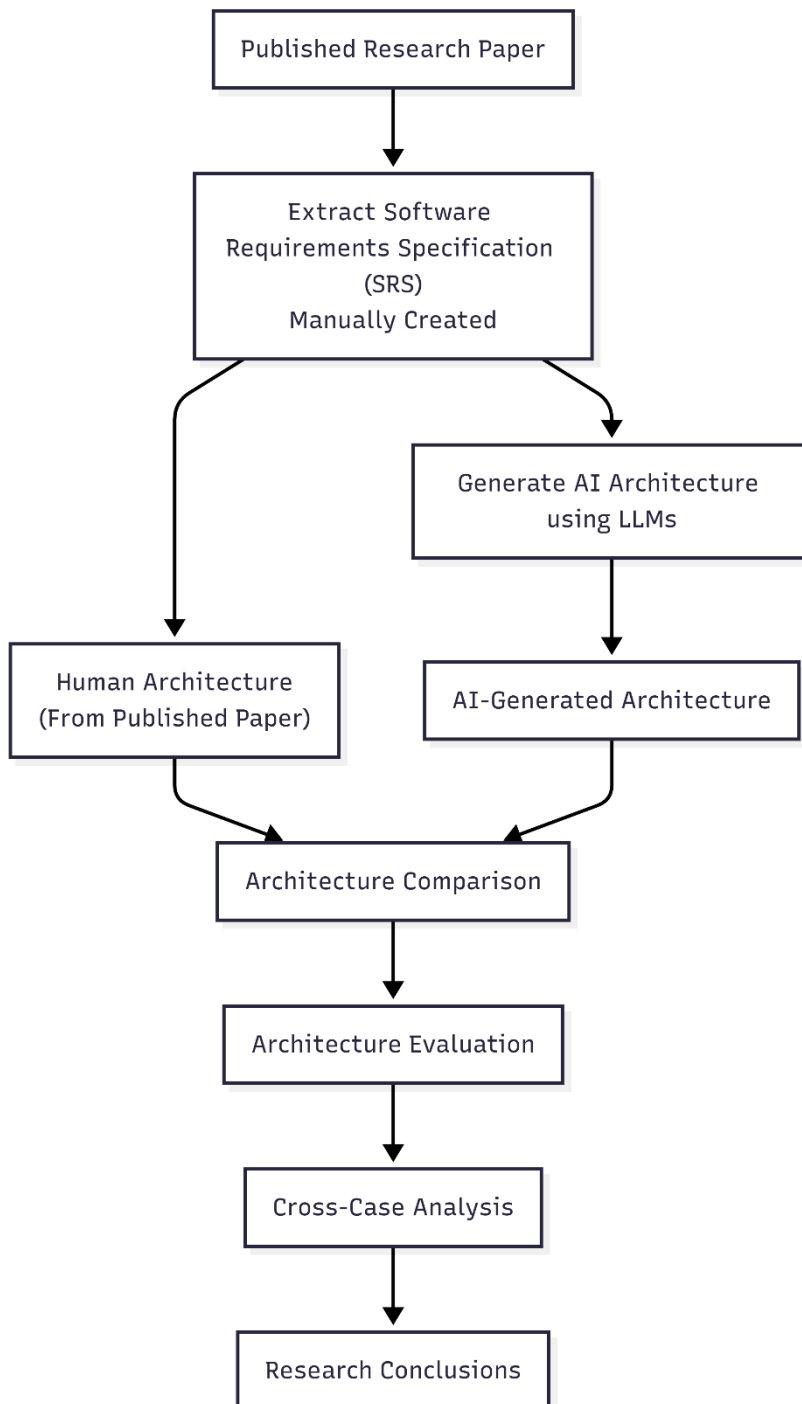
Comparative Evaluation of AI-Generated and Human-Designed Software Architectures

1. Research Objective

The practical work aims to evaluate whether AI-generated software architectures are comparable to human-designed software architectures when both are derived from the same Software Requirements Specification (SRS).

The study does not evaluate software implementation. Instead, it compares software architecture designs using a structured evaluation framework derived from software architecture literature.

2. Research Workflow



3. Practical Methodology

Phase 1 - Case Study Selection (Already Completed)

Select **three- five peer-reviewed software architecture papers**.

Selection criteria:

- Published in IEEE, Springer or ACM
- Contains a complete software architecture
- Contains sufficient system requirements
- Includes architecture diagrams
- Describes architectural decisions

Output:

- Case Study 1
- Case Study 2
- Case Study 3

Phase 2 - Requirements Extraction (Already Completed)

For each paper:

Extract a neutral Software Requirements Specification.

The SRS includes:

- System Overview
- Functional Requirements
- Non-functional Requirements
- Actors
- External Systems
- Business Rules
- Constraints

The original architecture is **not included** in the SRS to avoid bias during AI generation.

Output:

Case Study 1

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SRS

Phase 3 - AI Architecture Generation

For each SRS:

I have designed Zero short **prompt for generating architecture without any bias** .

Use the **different AI Models (ChatGPT, Gemini, Claude)**.

Use a **new conversation**.

Generate:

- Architecture Style
- Components
- Component Responsibilities
- Data Flow
- Deployment
- Mermaid Diagrams

Output:

SRS

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AI Architecture with reasoning

Phase 4 - Human Architecture Extraction (Already Present in Papers)

Extract from the paper:

- Architecture Style
- Components
- Layers
- External Systems
- Architecture Diagram

Output:

Paper

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Human Architecture

Phase 5 - Architecture Evaluation

Both architectures will be evaluated using the same framework.

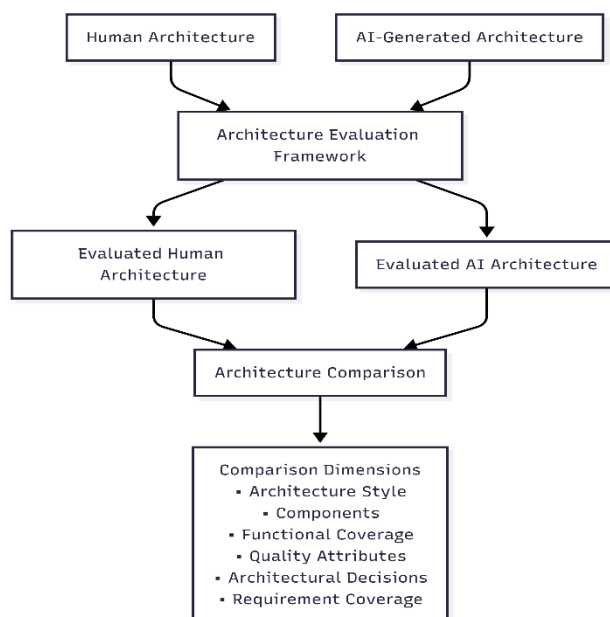
Criterion	Source
Functional Coverage	ISO/IEC 25010
Modularity	Bass et al.
Maintainability	ISO/IEC 25010
Scalability	Bass et al.
Interoperability	ISO/IEC 25010
Security	ISO/IEC 25010
Requirements-to-Architecture Traceability	ISO/IEC/IEEE 29148 + Software Architecture literature

Each criterion will be assessed using a predefined **1–5 rubric**.

Scores will be supported by architectural evidence rather than subjective opinion.

6. Comparison Process

For every case study:



7. Cross-Case Analysis

After evaluating all case studies:

Identify recurring patterns.

For example:

- Which architecture style does ChatGPT/ Gemini/ other LLMS prefer?
 - Which quality attributes are consistently satisfied?
 - Which components are commonly omitted?
 - Which architectural decisions differ from the human design?
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8. Expected Outputs

For each case study:

1. Original Paper
 2. Extracted SRS
 3. AI-generated Architecture
 4. Human Architecture Summary
 5. Evaluation Table
 6. Comparison Table
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Overall outputs:

Cross-case Analysis

Findings

Conclusions

9. Research Contribution

This study contributes:

- A reproducible methodology for comparing AI-generated and human-designed software architectures.
- A standardised process for generating software architectures from published software requirements.
- A structured evaluation framework based on established software architecture quality attributes.
- An empirical analysis of similarities and differences between AI-generated and human-designed architectures.

10. Research Questions (I am gonna answer)

RQ1) How do AI-generated software architectures compare with human-designed software architectures when derived from the same Software Requirements Specification (SRS)?

RQ2) To what extent do AI-generated software architectures satisfy the functional and quality requirements specified in the Software Requirements Specification?

RQ3) What similarities and differences exist between AI-generated and human-designed software architectures in terms of architectural structure, quality attributes, and requirements-to-architecture traceability?